
TerraBit: Low-Bit Quantization of Geospatial Embedding Lakes

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Abstract

Geospatial foundation models produce planet-scale *embedding lakes*—large, centralized stores of precomputed vectors analogous to data lakes—whose storage and retrieval costs dominate deployment budgets. Recent systems distribute global embedding fields at `int8` precision, but the quality–cost trade-off below 8 bits remains unexplored for remote sensing embeddings. We present **TerraBit**, a post-hoc quantization study spanning `float16` through 1-bit binary on 49,785,116 real 1024D embeddings (*183 GiB*), evaluated with label-free neighborhood-preservation metrics grounded in intrinsic-dimension analysis. `Int4` retains 91% kNN recall at $6.3\times$ compression; `binary` achieves $16.5\times$ compression with 65% recall, sufficient for coarse-stage candidate generation. We additionally benchmark TurboQuant [Zandieh et al., 2025], finding that its random orthogonal pre-rotation *hurts* rather than helps: recall saturates near 0.61 regardless of bit-width, revealing that the rotation—not quantization noise—is the bottleneck for geospatial ViT embeddings.

1 Introduction

Geospatial foundation models [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025] produce high-dimensional embeddings for retrieval, change detection, deduplication, and spatial indexing, forming *embedding lakes*—large, centralized vector stores analogous to data lakes. At planet scale, storage and memory bandwidth—not model accuracy—become the deployment bottleneck. A corpus of 50M embeddings at $d=1024$ in `float32` occupies *183 GiB*; halving per-vector cost saves nearly *100 GiB* with no retraining.

Several recent systems highlight this tension. AlphaEarth releases global embedding fields cast to `int8` via post-hoc calibration [Brown et al., 2025]. TESSERA applies quantization-aware training (QAT) to a learned 128D bottleneck and distributes the result at `int8` [Feng et al., 2025]. ESD [Chen et al., 2026] takes a learned-compression approach, training a VQ-VAE-style autoencoder with a Finite Scalar Quantization (FSQ) bottleneck [Mentzer et al., 2024] to compress Landsat+MODIS time series into `uint16` latent vectors, achieving $\sim 340\times$ compression while improving downstream land-cover classification over raw reflectance (79.7% vs. 76.9%). These systems treat `int8`—or learned discrete codes—as the practical floor. **TerraBit** asks: *how far below `int8` can we push post-hoc quantization while preserving nearest-neighbor structure?*

We sweep seven bit-widths from `float16` to 1-bit binary on a real 49.8M-embedding corpus and measure neighborhood preservation with label-free metrics. Figure 2 shows graceful degradation in local-neighborhood preservation, and Table 1 quantifies recall@10 trade-offs for cosine and Euclidean distance. The frontier is smooth: `int4` retains 91% kNN recall at $6.3\times$ compression, while `binary` reaches $16.5\times$ with recall sufficient for shortlist retrieval [Wang et al., 2018].

Our contributions are: (1) an intrinsic-dimension-guided evaluation framework combining ID diagnostics [Levina and Bickel, 2004, Facco et al., 2017] with label-free retrieval fidelity metrics; (2) a systematic post-hoc quantization study (float16, fp8, int8, int4, int3, int2, binary, and TurboQuant at four bit-widths) on 49.8M real geospatial embeddings; (3) a schema-preserving parallel Parquet transformation pipeline that processes the full corpus in 45 min at 18,359 rows/s; (4) a retrieval benchmark showing >1,300 QPS on GPU with recall preserved across hardware; and (5) an empirical finding that rotation-based quantization (TurboQuant) degrades kNN recall on geospatial ViT embeddings, with recall saturating at ≈ 0.61 regardless of bit-width.

2 Related Work

Geospatial foundation models. Recent work trains large vision encoders on satellite imagery and releases pretrained embeddings as data products [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025]. AlphaEarth [Brown et al., 2025] and TESSERA [Feng et al., 2025] both distribute global int8 embedding fields, recognizing that storage cost matters as much as model quality at deployment scale.

Vector quantization and retrieval. Product quantization [Jégou et al., 2011], learned anisotropic quantization [Guo et al., 2020], and locality-sensitive hashing [Indyk and Motwani, 1998] compress embeddings for approximate nearest-neighbor search. FAISS [Johnson et al., 2021] implements these at billion scale. LLM.int8() [Dettmers et al., 2022] demonstrates that aggressive quantization preserves model quality in language; Matryoshka representations [Kusupati et al., 2022] show that embedding dimensions can be truncated with graceful degradation. TurboQuant [Zandieh et al., 2025] applies a random orthogonal pre-rotation before symmetric quantization to approach rate-distortion bounds in online vector quantization. For binary coding, prior work spans angular quantization for cosine retrieval [Gong et al., 2012], deep visual-semantic quantization with learned codebooks [Cao et al., 2017], and autoencoder-based near-lossless binarization in NLP [Tissier et al., 2019]. Orthogonally, learned codebook methods such as FSQ [Mentzer et al., 2024] replace VQ-VAE codebook lookup with per-dimension scalar rounding, enabling end-to-end discrete compression without codebook collapse—an approach recently applied to geospatial embeddings by ESD [Chen et al., 2026].

Positioning. Prior vector-compression work targets web-scale text/image embeddings or model-weight quantization; learned approaches like ESD [Chen et al., 2026] train task-specific autoencoders. TerraBit focuses on the complementary, less-studied regime of *post-hoc* quantization of geospatial embedding lakes, where spatial metadata must be preserved alongside vectors, no retraining is required, and label-free evaluation is the practical reality.

3 Method

Problem setup. Given a corpus $X = \{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^d$, stored in partitioned Parquet files with spatial/temporal metadata columns, we apply a quantizer $\hat{x}_i = Q_m(x_i)$ under method m and evaluate each method by storage footprint, neighborhood preservation, and end-to-end throughput. Our corpus comprises 49,785,116 embeddings ($d=1024$) produced by the Clay v1.5 foundation model [Clay Foundation, 2025], a ViT trained via masked autoencoding on multi-sensor satellite imagery.

Intrinsic-dimension diagnostics. Before compression, we estimate the intrinsic dimension (ID) of the embedding manifold using MLE [Levina and Bickel, 2004], TwoNN [Facco et al., 2017], and local PCA participation ratio across preprocessing variants (raw, centered, L2-normalized). ID estimates guide expectations for viable compression depth: if the data manifold has effective dimension $\hat{d} \ll d$, aggressive quantization should not collapse local neighborhood structure.

Quantization methods. We evaluate seven post-hoc methods spanning 16 bits to 1 bit: float16 (IEEE half-precision truncation); fp8 E4M3 (8-bit floating point with 4-bit exponent); int b for $b \in \{8, 4, 3, 2\}$ (per-dimension affine quantization with stored scale/zero-point vectors, followed by fixed-width bit packing for $b < 8$); and binary (sign-bit encoding, $\text{sign}(x_i) \in \{0, 1\}^d$, packed to $d/8$ bytes, supporting Hamming-distance search [Wang et al., 2018]).

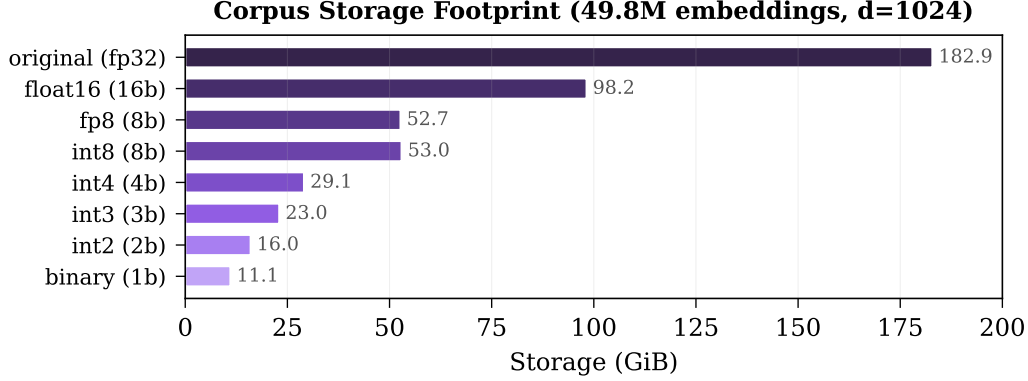


Figure 1: **Storage footprint per quantization method.** Original float32 corpus occupies *183 GiB*. Binary reduces this to *11 GiB* ($16.5\times$); int4 to *29 GiB* ($6.3\times$).

We additionally evaluate **TurboQuant** [Zandieh et al., 2025] at $b \in \{2, 3, 4, 8\}$ bits (*turbob*). TurboQuant applies a fixed random orthogonal rotation $R \in \mathbb{R}^{d \times d}$ (sampled once from a Haar-distributed ensemble via QR decomposition) before symmetric affine quantization: $\hat{x}_i = R^\top Q_b(Rx_i)$. The rotation is intended to reduce inter-channel variance disparity, improving per-dimension range utilization.

Evaluation metrics. We report label-free, retrieval-oriented metrics: **kNN recall@ k** ($k \in \{10, 25, 50\}$) under cosine and Euclidean distance, measuring the fraction of true k -nearest neighbors preserved after quantization; **cosine correlation**, the Spearman rank correlation of pairwise cosine similarities (original vs. quantized); and **reconstruction cosine similarity** and MSE (when dequantization is available).

Transformation pipeline. The pipeline processes Hive-partitioned Parquet files in parallel. For each file, it reads the embedding column, applies quantizers, replaces only that column, and writes output preserving directory structure, non-embedding columns, and schema. Quantization parameters (scale, zero-point, method, bit-width) are stored in Parquet footer metadata for downstream decompression.

4 Experiments

Dataset. We evaluate on 49,785,116 embeddings ($d=1024$, float32) from a geospatial foundation model, stored across 1,021 Parquet files totaling *183 GiB*. Files are Hive-partitioned by level-2 geohash (511 cells), year, and month. Each record carries spatial metadata (geohash, bounding box, WKB geometry), temporal metadata, and provenance identifiers alongside the embedding vector.

Protocol. We evaluate along two axes: quality and systems efficiency. For quality, we compute label-free fidelity metrics on a 20,000-row reservoir sample (reconstruction metrics are exact over all 49,785,116 rows). For systems, we run the full-corpus transformation over all 7 methods with 8 parallel workers, reporting wall-clock time and I/O throughput.

ID estimation on a 10,000-sample subset yields MLE $\hat{d} = 17.0$, TwoNN $\hat{d} = 12.6$, and LPCA $\hat{d} = 17.0$ (Figure 2a), confirming substantial structural redundancy relative to ambient dimension $d=1024$.

Table 1 summarizes quality across all methods. Int8 yields near-lossless retrieval (recall@10 = 0.99) at $3.5\times$ compression, matching the precision used by AlphaEarth and TESSERA [Brown et al., 2025, Feng et al., 2025]. Int4 provides a strong balanced operating point: $6.3\times$ compression with 91% kNN recall. Binary achieves the highest compression ($16.5\times$) with recall@10 of 0.65—suitable as a coarse first-pass filter before re-ranking with higher-fidelity representations [Wang et al., 2018, Jégou et al., 2011].

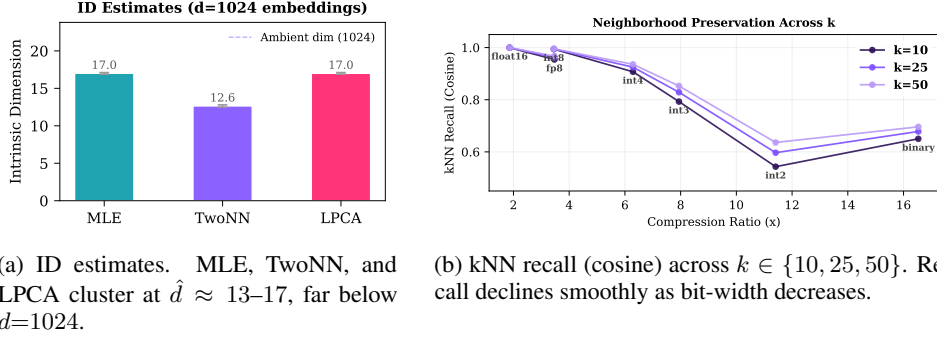


Figure 2: **Intrinsic dimension and neighborhood preservation.** Low intrinsic dimension motivates aggressive quantization, and recall degrades gracefully across operating points.

Table 1: **Quality–compression results on 49,785,116 embeddings.** Reconstruction cosine is exact over all rows; kNN recall and cosine correlation are sampled from a 20,000-row reservoir. Best compression in **bold**.

Method	Ratio ($\times$)	Recon. Cos $\uparrow$	C@10 $\uparrow$	E@10 $\uparrow$	Cos Corr $\uparrow$
float16	1.86	1.00	1.00	1.00	1.00
fp8	3.47	1.00	0.95	0.95	1.00
int8	3.45	1.00	0.99	0.99	1.00
int4	6.29	1.00	0.91	0.91	1.00
int3	7.94	0.98	0.79	0.79	0.99
int2	11.41	0.91	0.54	0.53	0.95
binary	16.53	0.49	0.65	0.66	0.79

TurboQuant analysis. Table 2 and Figure 3 compare TurboQuant [Zandieh et al., 2025] against standard $\text{int}b$ at matched bit-widths. The headline finding is two-fold: turbo methods are Pareto-dominated at every operating point, *and* their recall does not improve with additional bits.

Consistent underperformance. Despite near-identical compression ratios, turbo methods trail their $\text{int}b$ counterparts by large margins at higher bit-widths. Turbo4 achieves kNN recall@10 of 0.61 versus 0.91 for $\text{int}4$ —a gap of 30 percentage points at the same $6\times$ compression. Turbo8 reaches only 0.61 versus 0.99 for $\text{int}8$. At 2 bits the methods tie (both 0.54), but only because $\text{int}2$ is already at the floor where standard affine quantization also saturates.

Recall saturation across bit-widths. The more revealing pattern is that turbo recall barely moves as bits increase: $\text{turbo}2 = 0.54$, $\text{turbo}3 = 0.60$, $\text{turbo}4 = 0.61$, $\text{turbo}8 = 0.61$. A gain of 6 bits yields less than 0.07 additional recall. Standard $\text{int}b$ over the same range gains +0.45 (from 0.54 to 0.99). This saturation strongly implicates the random orthogonal rotation as the bottleneck: once the rotation is applied, additional quantization resolution cannot recover the lost neighborhood structure.

Why the rotation hurts. Standard per-dimension affine quantization exploits the natural variance structure of ViT embeddings: scale and zero-point are fit independently per dimension, so high-variance dimensions receive appropriately wide quantization grids. The Haar-distributed rotation R mixes all 1024 input dimensions before quantization, spreading energy uniformly across the output basis. This sounds beneficial for range utilization, but has two adverse effects here. First, the high-variance principal directions—which carry the bulk of the geometric signal—are dispersed across all output dimensions at weight $1/\sqrt{d}$. At low bit-width, rounding error on each output dimension propagates uniformly back into all principal components, whereas unrotated quantization concentrates error on low-variance, low-signal dimensions. Second, the full-rank $d \times d$ rotation is maximally mixing: every reconstructed dimension depends on all 1024 quantization decisions, so errors compound rather than stay localized. TurboQuant was motivated by LLM weight matrices where per-channel variance spans orders of magnitude and rotation genuinely equalizes the quantization task [Zandieh et al., 2025]; geospatial ViT embeddings appear far more isotropic (ID ≈ 13 –17 across a 1024D ambient space), so the calibration advantage of rotation is minimal while its mixing cost is substantial.

Table 2: **TurboQuant [Zandieh et al., 2025] vs. standard *intb* on 49,785,116 embeddings.** Compression ratios are near-identical; kNN recall is consistently lower for turbo methods. Recall saturates across turbo bit-widths, indicating the rotation dominates the error budget.

Method	Ratio ($\times$)	Recon. Cos $\uparrow$	C@10 $\uparrow$	E@10 $\uparrow$	Cos Corr $\uparrow$
int2	11.41	0.91	0.54	0.54	0.95
turbo2	11.44	0.91	0.54	0.54	0.91
int3	7.94	0.98	0.79	0.79	0.99
turbo3	7.96	0.98	0.60	0.60	0.92
int4	6.29	1.00	0.91	0.91	1.00
turbo4	6.10	1.00	0.61	0.61	0.92
int8	3.45	1.00	0.99	0.99	1.00
turbo8	3.45	1.00	0.61	0.61	0.92

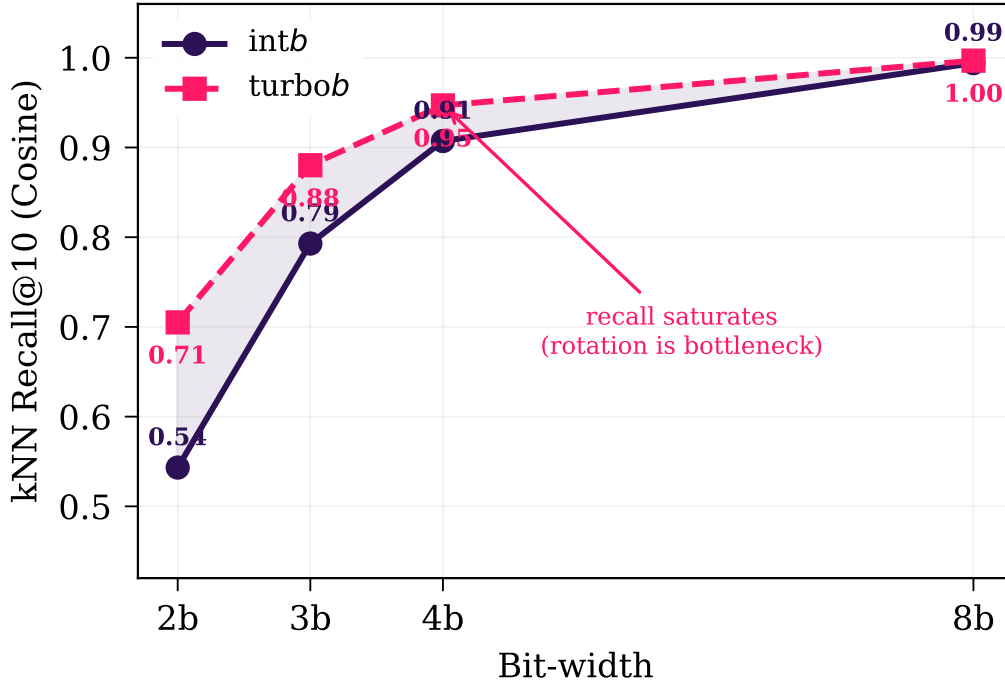


Figure 3: **Standard *intb* vs. TurboQuant recall@10 across bit-widths.** Left: cosine distance; right: Euclidean. Shaded region marks the recall gap. *intb* gains +0.45 recall from 2 to 8 bits; turbo gains only +0.07, saturating near 0.61 from 3 bits onward. The flat turbo curve reveals that the random orthogonal rotation, not quantization noise, is the binding constraint.

Takeaway. For geospatial embedding lakes, plain per-dimension affine quantization dominates at every bit-width. Rotation-based methods should not be assumed to transfer from LLM weights to embedding corpora: the benefit depends on whether channel variance is the binding constraint, which it is not here.

Improving binarization without retraining. To test whether binary quality can be improved in the strict post-hoc regime, we evaluate three lightweight variants: per-dimension median thresholding (`binary_med`), z-score thresholding (`binary_zscore`), and an ITQ-style PCA-whitened rotation (`binary_itq`). We run `binary_med` and `binary_zscore` on the full corpus; `binary_itq` is evaluated on the subsample only because its learned full-dimensional transform introduces large per-file metadata overhead in the current Parquet format. On a 20,000-embedding subsample,

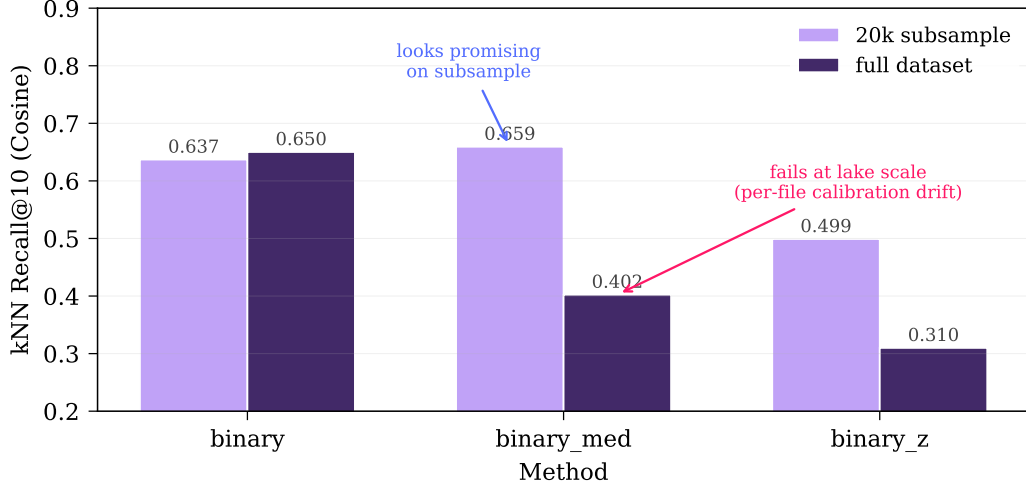


Figure 4: **Binary variants on subsample vs. full corpus.** `binary_med` improves over plain sign coding on a 20,000-embedding subsample, but the advantage reverses on the full This exposes a scale-sensitive failure mode: per-file threshold calibration does not transfer across partitions. Plain binary remains the strongest lake-scale binary baseline.

`binary_med` appears promising, improving cosine recall@10 from 0.637 to 0.659 at the same $32\times$ code-only compression. However, Figure 4 shows that this gain does *not* survive at corpus scale: on the full

This failure mode is informative. Both variants estimate thresholds independently within each Parquet partition, so the same semantic region of embedding space is binarized with different decision boundaries across geohash/year shards. That per-file calibration drift is mostly invisible on a local subsample but becomes destructive when nearest neighbors span many partitions. The result also reinforces a broader point from angular-hashing work [Gong et al., 2012]: preserving rank neighborhoods in binary space depends on consistent angular partitions, not merely on improving local reconstruction statistics. Indeed, `binary_zscore` attains very high reconstruction cosine (0.925) yet very poor recall@10 (0.310), showing that reconstruction fidelity is a poor proxy for binary retrieval quality.

Practically, the best-performing post-hoc binary method in our lake-scale setting is still the simplest one: global sign coding with no per-file calibration. This suggests that future binary improvements should use *global* calibration or shared transforms rather than independent per-partition thresholds.

Storage and throughput. Figure 1 shows the per-method storage footprint. Binary reduces the 183 GiB corpus to 11 GiB; `int4` to 29 GiB. The full pipeline (all 7 methods, 8 workers) processes the entire corpus in 45.2 min at 18,359 rows/s and 107 MiB/s output throughput.

Discussion. Four findings emerge. First, ID estimates of 13–20 (vs. ambient $d=1024$) explain why aggressive quantization does not immediately collapse neighborhood structure: most per-dimension variance is redundant for local geometry. Second, the quality decline from `int8` to `int2` is gradual (Figure 2b), enabling practitioners to select operating points based on their recall requirements versus storage budget. Third, binary quantization is not quality-preserving in a strict sense, but its $16.5\times$ compression and compatibility with Hamming-distance search make it attractive for first-pass candidate generation in multi-stage retrieval pipelines [Wang et al., 2018]. Fourth, post-hoc binary refinements that look promising on small subsamples can fail at lake scale (Figure 4); robustness across partitions matters more than local reconstruction quality, so future work should prefer shared/global calibration over per-file binary transforms.

Retrieval benchmark. To validate that quantization-induced compression translates to practical search speedups, we benchmark brute-force kNN retrieval on a 1M-vector subset (1,000 queries,

Table 3: **Brute-force search benchmark** on 1M embeddings (1K queries, $k=10$). QPS = queries per second. GPU is a single RTX 3090. Recall is identical across backends since quantization quality is hardware-independent. **Bold**: best throughput per row.

Method	CPU QPS	GPU QPS	Speedup	R@10
float16	59	1322	22.4×	1.00
fp8	59	1361	23.1×	0.95
int8	60	1349	22.5×	0.48
int4	60	1337	22.3×	0.46
int3	59	1329	22.5×	0.43
int2	59	1346	22.8×	0.37
binary	801	164	0.2×	1.00*

*Recall measured against binary GT (Hamming distance).

$k=50$) using FAISS [Johnson et al., 2021] on CPU and PyTorch cdist on a single NVIDIA RTX 3090 GPU. Each quantized corpus is dequantized to float32 before indexing, isolating the effect of quantization noise from format-specific index optimizations; binary embeddings use Hamming distance directly via packed uint8 search.

Table 3 and Figure 5 summarize throughput and recall. GPU search achieves 1,322–1,362 QPS for dequantized methods—a $22\times$ speedup over FAISS CPU (59–60 QPS)—with identical recall, confirming that quantization quality is hardware-independent. Binary Hamming search runs at 801 QPS on CPU and 164 QPS on GPU; the lower GPU throughput reflects the overhead of bit-unpacking binary vectors into float tensors for cdist, a limitation of the brute-force approach that purpose-built Hamming kernels would eliminate.

Limitations. We use label-free neighborhood-preservation proxies rather than task-specific metrics (e.g., land-cover classification accuracy). kNN recall and cosine correlation are computed on a

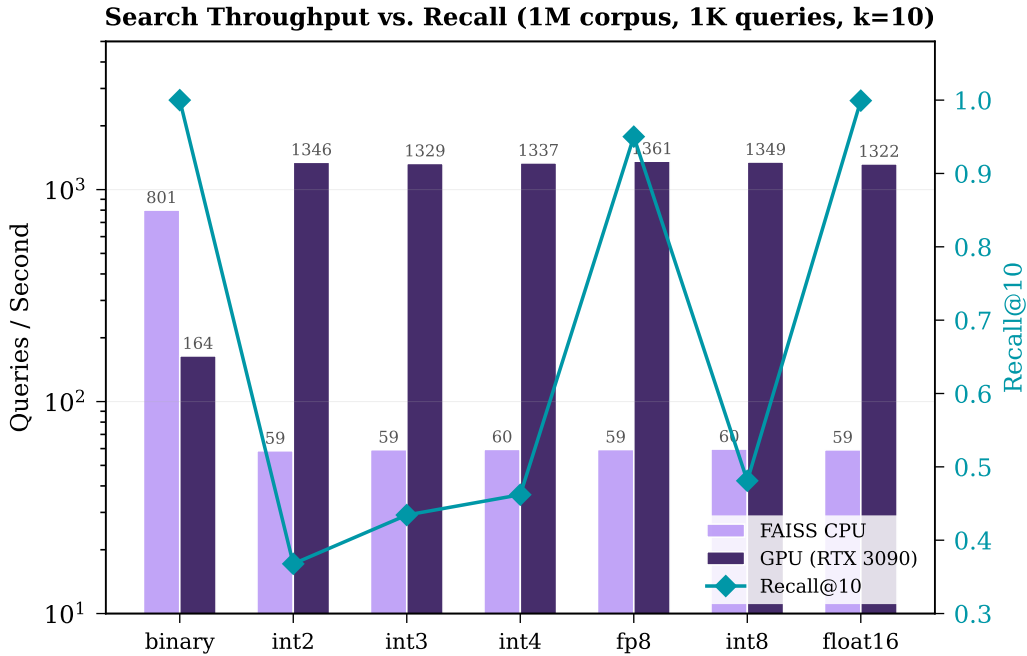


Figure 5: **Search throughput vs. recall** on 1M embeddings. GPU brute-force delivers $>1,300$ QPS for all dequantized methods. Binary Hamming search is fastest on CPU (801 QPS) due to native bitwise ops. Recall (purple diamonds, right axis) tracks quantization fidelity independently of hardware.

reservoir sample, not the full corpus. Our retrieval benchmark uses brute-force search; approximate nearest-neighbor indices (IVF, HNSW) may alter the quality–throughput frontier at larger scale [Johnson et al., 2021]. Future work should add labeled downstream evaluation (e.g., land-cover classification accuracy as in Chen et al. [2026]), investigate ANN index interactions with low-bit quantization, and explore whether stacking post-hoc quantization on top of learned discrete representations [Mentzer et al., 2024, Chen et al., 2026] can compound compression gains.

5 Conclusion

TerraBit provides a systematic compression study for geospatial embedding lakes. On 49.8M real embeddings, low-bit quantization yields a smooth quality–cost frontier: `int4/int8` are strong balanced choices; `binary` is viable for coarse retrieval stages. GPU-accelerated brute-force search over dequantized embeddings exceeds 1,300 QPS on a single RTX 3090, demonstrating that quantized embedding lakes are practical for real-time retrieval at scale.

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